GATE SOLUTION
IN
MINING ENGINEERING

(Containing Solved Questions 2015 - 2007)

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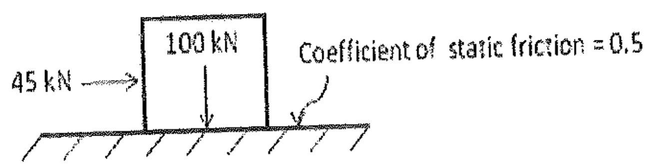
Gate Solution-2014

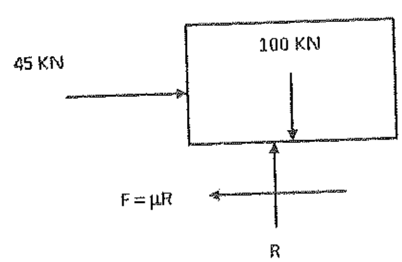
Q.1 A block of weight 100 kN rests on a floor as shown in the figure. The coefficient of static friction between the block and the floor is 0.5. A force of 45 kN is applied horizontally on the block. The static frictional force in kN is



(A) 22.5 (B) 50.0 (C) 55.0 (D) 100.0

Answer (B)

Solution.



Where , F - is frictional force

R – Reaction force

$\mu = 0.5$ = Coefficient of static friction

Now $\sum V = 0$

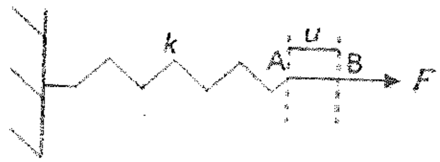
$R - 100 = 0$

$R = 100KN$

Hence , frictional force (F) , $F = \mu R$

$$F = 0.5 \times 100 = 50\ N$$

Q.2 A spring of constant stiffness k is stretched from point A to point B (displacement u in the figure) by a force F. The potential energy of the spring is expressed by



(A) $\frac{1}{2} ku - Fu$ (B) $\frac{1}{2} ku + Fu$

(C) $ku - F$ (D) $ku + F$

Answer (A)

Solution.

Potential Energy of spring is given by = $\frac{1}{2}Kx^2$

Conservative force – Conservative force is defined as one that does mechanical work independent of path of motion and such that the work is reversible or recoverable.

Therefore, in case of spring, the mechanical work of a conservative force is considered to be loss in potential energy of spring and that is : -

$$U_F = -W = -Fu$$

Hence, potential Energy (P_s) of spring is given by

$$P_s = \frac{1}{2}Kx^2 - Fu$$

Q.3 If σ_s is the induced stress and σ_i is the insitu stress at a point below ground, the 'stress concentration' at that point is

(A) $\sqrt{\frac{\sigma_s}{\sigma_i}}$ (B) $\sqrt{\frac{\sigma_i}{\sigma_s}}$ (C) $\frac{\sigma_i}{\sigma_s}$ (D) $\frac{\sigma_s}{\sigma_i}$

Answer (D)

Q.4 The components of state of stress at a point in x–y plane are given as σ_{xx} =5 MPa, σ_{yy}= 10MPa and τ_{xy} = – 2MPa. The sum of the principal stresses acting on the x–y plane in MPa is

____

Answer : 15 to 15.

Solution.

Given that

σ_{xx} = 5 MPa

σ_{yy} = 10 MPa

τ_{xy} = – 2 MPa

Major principal stress is given by : -

$$\sigma_1 = \frac{1}{2}\left(\sigma_{xx} + \sigma_{yy}\right) + \left\{\frac{1}{4}\left(\sigma_{xx} - \sigma_{yy}\right)^2 + \tau_{xy}^2\right\}^{1/2}$$

$$\sigma_1 = \frac{1}{2}(5+10) + \left\{\frac{1}{4}(5-10)^2 + (-2)^2\right\}^{1/2}$$

σ_1 = 10.7015 MPa.

and minor principal stress is given by :-

$$\sigma_2 = \frac{1}{2}\left(\sigma_{xx} + \sigma_{yy}\right) - \left\{\frac{1}{4}\left(\sigma_{xx} - \sigma_{yy}\right)^2 + \tau_{xy}^2\right\}^{1/2}$$

$$\sigma_2 = \frac{1}{2}(5+10) - \left\{\frac{1}{4}(5-10)^2 + (-2)^2\right\}^{1/2}$$

$$\sigma_2 = 4.30 \text{ MPa}$$

Hence, Sum of principal stresses is given by

$$\Sigma\sigma = \sigma_1 + \sigma_2$$

$$= 10.70 + 4.30$$

$$= 15 \text{ MPa}$$

Q.5 The angle $5^\circ 15' 25''$ is expressed in hours, minutes, and seconds as

(A) $1^h 20^{min} 1.67^s$ **(B) $1^h 20^{min} 16.00^s$**

(C) $0^h 21^{min} 1.67^{min}$ **(D) $0^h 21^{min} 16.00^s$**

Answer (C)

Solution.

We know, $1^o = \frac{1}{15} h$

$$5^o = \frac{5}{15} h$$

$$5^o = 0^h 20^m 0^S$$

and $1' = \frac{1}{15}\mathrm{m}$

$$15' = \frac{15}{15}\mathrm{m}$$

$$15' = 0^h 1^m 0^S$$

and $$1'' = \frac{1}{15} s$$

$$25'' = \frac{25}{15} s$$

$$25'' = 0^h 0^m 1.67^S$$

Hence, Angle $5^0 15' 25'' = 0^h 21^m 1.67^s$

Q. 6 **A circular curve has a radius of 200 m and deflection angle of 65^0. The length of the curve in m is**

(A) 221 **(B) 227** **(C) 235** **(D) 262**

Answer (B)

Solution.

We know,

$$\text{Angle} = \frac{Arc}{radius}$$

Or Length of curve = Angle x radius

$$= \frac{\pi}{180^0} \times 65^0 \times 200$$

$$= 227 \text{ m}$$

Q.7 **The weight strength of ANFO of specific gravity 0.8 is 912 kcal/kg. The weight strength of an emulsion explosive of specific gravity 1.2 is 850 kcal/kg. Bulk strength of the emulsion explosive relative to ANFO in percentage is**

Answer : 135 to 142.

Solution.

$$\text{Relative bulk strength} = \frac{\text{Relative weight strength of explosive} \times S.G.\ \text{of explosive}}{\text{Relative weight strength of ANFO} \times S.G.\ \text{of ANFO}}$$

So, Bulk strength of emulsion explosive $= \dfrac{850 \times 1.2}{912 \times 0.8}$

$= 1.40$

$= 140\%$

Q.8 **In a cut-and-fill stope, the main purpose of back filling is to**

(A) reduce ore dilution

(B) prevent high stress concentrations in far field domain

(C) prevent displacement due to dilation of fractured wall rock

(D) improve ore rehandling

Answer (C)

Q.9 **Bypass valve in a compressed oxygen type self-contained breathing apparatus is meant to**

(A) release accumulated nitrogen in the breathing bag

(B) release excess pressure in the breathing bag

(C) supply oxygen directly to wearer in case pressure reducing valve does not function

(D) flush out the apparatus with oxygen on opening the cylinder valve

Answer (C)

Q.10 **Given S is the setting load and Y is the yield load of a hydraulic prop, the correct relationship is**

(A) S < Y **(B) S > Y** **(C) S = Y** **(D) S = Y^2**

Answer (A)

Q.11 **Solution of the differential equation $dy/dx = ky$ follows exponential decay (where k is a constant) for $x \in [0, \infty]$ if**

(A) $k > 0$ **(B) $k < 0$** **(C) $k = 0$** **(D) $k = e$**

Answer (B)

Solution.

Note – Quantity is subjected to exponential decay if it decreases at a rate proportional to its current value. This can be expressed by differential equation where y is quantity and K is exponential decay constant.

$$\frac{dy}{dx} = -ky \text{ for } x \in [0, \infty]$$

but given differential equation

$$\frac{dy}{dx} = Ky \text{, so here } K < 0$$

Q.12 **The value of k for which the vectors $a = 2i - 3j$ and $b = ki + 4j$ are orthogonal to each other is ________**

Answer : 6 to 6.

Solution.

Given vectors

$$a = 2i - 3j$$

$$b = Ki + 4j$$

when two vectors are orthogonal to each other then,

$$a.b = 0$$

$$(2i - 3j) . (Ki + 4j) = 0$$

$$2K - 12 = 0$$

$$\therefore \quad K = 6$$

Q.13 **Which one of the following is the most likely mode of slope failure for waste dump**

(A) Circular **(B) Wedge** **(C) Plane** **(D) Toppling**

Answer (A)

Q.14 **The occurrence of head in a single toss of an unbiased coin is given by a random variable X. The variance of X is ____________**

Answer : 0.25 to 0.25.

Solution.

The occurrence of head in single toss of an unbiased coin = $\frac{1}{2}$

So, random variable = $\frac{1}{2}$.

Hence, variance of random variable X = $\left(\frac{1}{2}\right)^2$

$$= \frac{1}{4} = 0.25$$

Q.15 **The divergence of the vector $v = (x + y)(-yi + xj)$ is**

(A) $y - x$ **(B) $x - y$** **(C) $x^2 - y^2$** **(D) $y^2 - x^2$**

Answer (B)

Solution.

Given vector is, $v = (x + y)(-yi + xj)$

We know,

Divergence $F = \nabla.F = \nabla.V$

and $\nabla = I.\frac{\partial}{\partial x} + J.\frac{\partial}{\partial y} + K\frac{\partial}{\partial z}$

$$\text{div. } F = \nabla.V = \left(I.\frac{\partial}{\partial x} + J.\frac{\partial}{\partial y} + K\frac{\partial}{\partial z} \right).(x+y)(-yi+xj)$$

$$\text{div. } F = \left(I\frac{\partial}{\partial x} + J\frac{\partial}{\partial y} + K\frac{\partial}{\partial z} \right).\left\{-\left(xy+y^2\right)i + \left(x^2+xy\right)j\right\}$$

$$\text{div. } F = -\frac{\partial}{\partial x}\left(xy+y^2\right) + \frac{\partial}{\partial y}\left(x^2+xy\right)$$

$$\text{div. } F = -y + x$$

$$\text{div. } F = x - y$$

Q.16 **The value of $\lim_{x\to 0}\frac{|x|}{x}$ is**

(A) −1 **(B) 0** **(C) 1** **(D) non-existent**

Answer (D).

Solution.

Given function, $f(x) = \lim\limits_{x\to 0}\frac{|x|}{x}$

Putting $x = 0+h$, when $x \to 0$ then $h \to 0$

$$Rf(0+h) = \lim_{h\to 0}\frac{|0+h|}{0+h}$$

$$= \lim_{h \to 0} \frac{|h|}{h}$$

$$= \lim_{h \to 0} \frac{h}{h}$$

$= 1.$

Now putting $x = 0 - h$, when $x \to 0$, then $h \to 0$

$$Lf(0-h) = \lim_{h \to 0} \frac{|0-h|}{0-h}$$

$$= \lim_{h \to 0} \frac{h}{-h} = -1$$

Hence, $Rf\ (0+h) \neq Lf\ (0-h)$

Therefore , given function is not existent.

For Indian coal mines, the 'maximum allowable concentration' of respirable dust containing 7.5% free silica in mg/m^3 is

(A) 2.0 **(B) 2.2** **(C) 2.5** **(D) 2.7**

Answer (A)

ion.

We know

$$\text{Threshold limit} = \frac{15}{\%\,\text{respirable silica}}\,\text{mg}/\text{m}^3$$

$$= \frac{15}{7.5}\,\text{mg}/\text{m}^3$$

$$= 2\text{mg}/\text{m}^3$$

Q.18 Given k is the thermal conductivity, ρ is density and c is specific heat of a rock sample, the thermal diffusivity of the rock sample is

(A) $\frac{k\rho}{c}$ (B) $\frac{\rho c}{k}$ (C) $\frac{kc}{\rho}$ (D) $\frac{k}{\rho c}$

Answer (D)

Q.19 Cyclone, bag filter and scrubber can be used for control of

(A) water pollution (B) air pollution

(C) soil pollution (D) noise pollution

Answer (B)

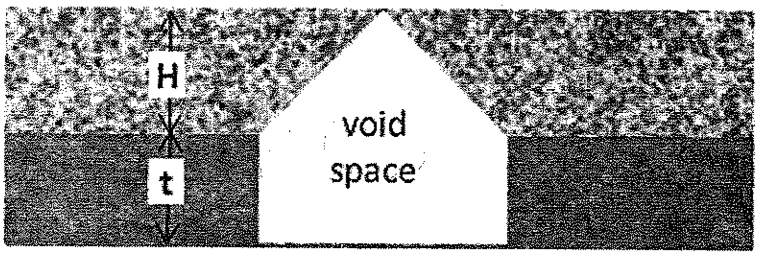
Q.20 A mine waste dump of pH 5.2 can be neutralized by adding

(A) urea (B) calcium carbonate (C) sulphuric acid (D) sodium chloride

Answer (B)

Q.21 A flat coal seam of thickness (t) 3 m is excavated and broken roof rock has completely filled the space created due to extraction as shown in the figure. If the bulking factor of roof rock is 1.2, the caving height (H) in m is ______________

Answer : 30 to 30.

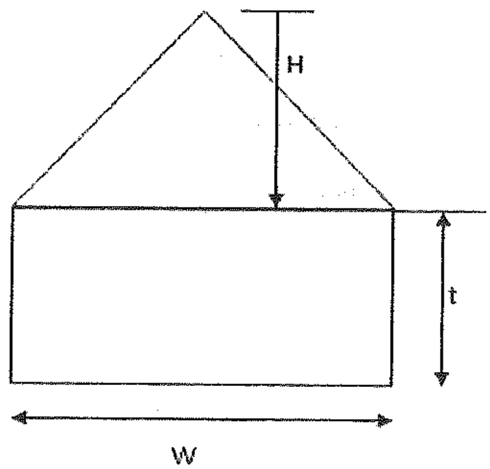


Solution.

Given that,

Coal thickness (t) = 3m

Bulking factor of root rock = 1.2



Bulking factor of rock = $\frac{Volume\ of\ rock\ after\ blasting}{Volume\ of\ rock\ before\ blasting}$

$$\text{Total area} = w \times t + \frac{1}{2} \times w \times H$$

Now,

$$1.2 = \frac{\left(w \times t + \frac{1}{2} \times w \times H\right) \times 1}{\frac{1}{2} \times w \times H \times 1}$$

$$1.2 \times \frac{1}{2} \times w \times H = \left(t + \frac{1}{2} \times H\right) . w$$

$$H\left(\frac{1.2}{2} - \frac{1}{2}\right) = 3$$

$$\therefore \quad H = \frac{3}{0.1} = 30 \text{ m}$$

Q.22 **A piece of coal sample weighs 10 kg in air and 2 kg when immersed in water. The specific**

gravity of the coal sample is ________

Answer : 1.25 to 125.

Solution.

Specific gravity = $\frac{W_1}{W_1 - W_2}$

Where,

W_1 – weight of solid in air =10 kg

W_2 – weight of solid in water = 2 kg

Hence, specific gravity of coal sample is = $\frac{10}{10-2}$

= 1.25

Q.23 **In a borehole log of 1.2 m in length, recovery of rock cores in cm is given below**

20, 8, 15, 8, 8, 4, 3, 9, 10, 1, 5, 10

The RQD in percentage is

(A) 29.2 **(B) 31.8** **(C) 45.8** **(D) 50.0**

Answer (C)

Solution.

We know,

$$RQD = \frac{\sum Length\ (L) of\ core\ pieces \geq 10\ cm\ length}{Total\ length\ of\ core\ run} \times 100$$

$$RQD = \frac{(20+15+10+10)\ cm}{1.2\ m} \times 100$$

$$RQD = \frac{(0.2 + 0.15 + 0.1 + 0.1)\ m}{1.2\ m} \times 100$$

RQD = 45.8%

Q.24 **An underground coal mine panel produces 520 tonnes per day deploying 220, 200 and 192 persons in three shifts. As per CMR 1957, the minimum quantity of air in m^3/min to be delivered at the last ventilation connection of the panel is ___________**

Answer : 1320 to 1320.

Solution.

Production per day = 520 tonnes

Total employee per day = 220 + 200 + 192 = 612

Here,

Total employee per day > Production per day

So,

Minimum quantity of air to be delivered shall be according to employees per day.

Hence,

Minimum quantity of air in $\mathrm{m}^3/\mathrm{min}$ to be delivered at the last ventilation connection of the panel is = number of workers in largest shift x 6 , m^3/min

= 220 x 6

= 1320 m^3/min

Q.25 **In a PERT network, the activities on the critical path are a, b and c. The standard deviations of the durations of these activities are 2, 2 and 1 respectively. The variance of the project duration is**

(A) 3 **(B) 5** **(C) 9** **(D) 12**

Answer (C)

Solution.

Given ,

$$\sigma_a = 2\ , \quad \sigma_b = 2\ , \quad \sigma_c = 1$$

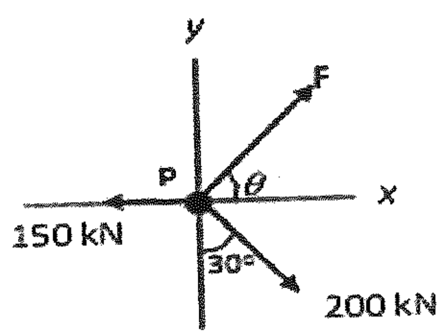
In PERT network,

$$(variance)_{abc} = (variance)_a + (variance)_b + (variance)_c$$

$$(variance)_{abc} = (\sigma^2)_a + (\sigma^2)_b + (\sigma^2)_c$$

$$(variance)_{abc} = 2^2 + 2^2 + 1^2 = 9$$

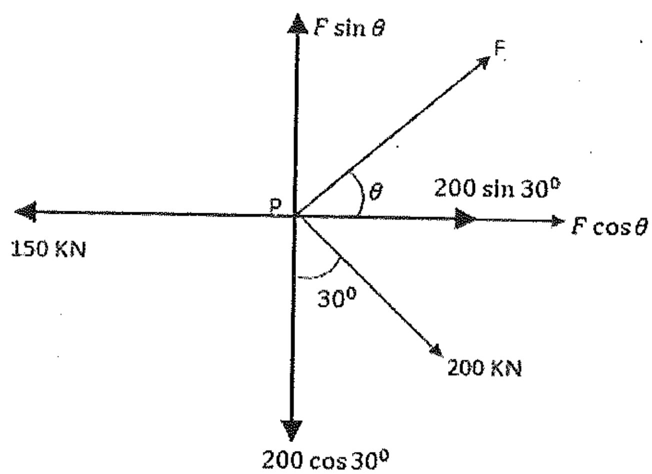
Q.26 A particle P is in equilibrium as shown in the figure. The magnitude in kN and the orientation θ in degrees of the force F respectively are



(A) 52.1, 16.1 (B) 221.2, 23.2 (C) 102.3, 53.4 (D) 180.3, 73.9

Answer (D)

Solution.



Now , $\sum H = 0$

$$-150 + F\cos\theta + 200\sin 30^0 = 0$$

$$F\cos\theta = 50 \text{ ....(i)}$$

And , $\sum V = 0$

$$F\sin\theta - 200\cos 30^0 = 0$$

$$F\sin\theta = 173.2050 \text{ ....(ii)}$$

From equation (i) & (ii), it is found that,

$$\tan\theta = \frac{173.2050}{50}$$

$$\tan\theta = 3.4641$$

$$\theta = \tan^{-1}(3.4641)$$

$$\theta = 73.9^0$$

From equation (i),

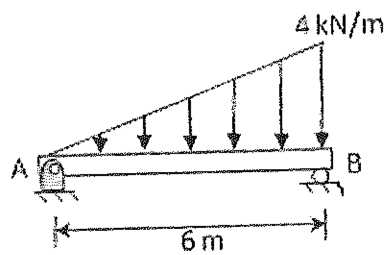
$$F\cos\theta = 50$$

$$F\cos 73.9^0 = 50$$

$$F = \frac{50}{\cos 73.9^0}$$

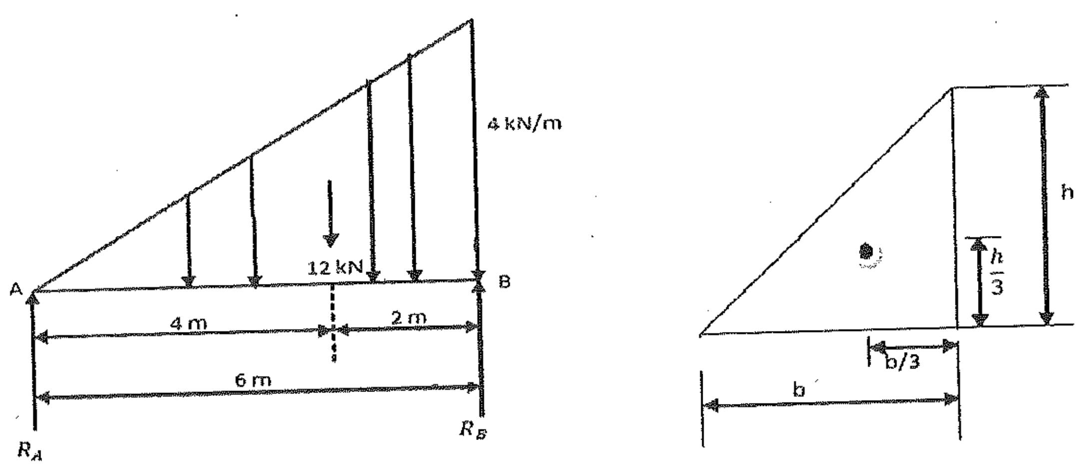
F = 180.3KN

Q.27 **A distributed load of 4 kN/m acts on a beam of 6 m length supported by a hinge and a roller as shown in the figure. The distance in m of the point of zero shear in the beam from the point A is _____.**



Answer : 3.4 to 3.5.

Solution.



Now , $\sum V = 0$

$$R_A + R_B - \frac{1}{2} \times 6 \times 4 = 0$$

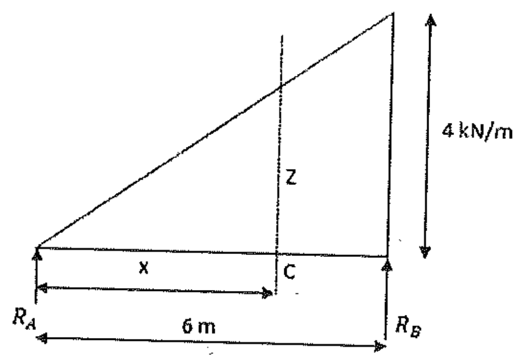
$$R_A + R_B = 12\ kN \text{ ....(i)}$$

Taking moment about A,

$$R_B \times 6 - \frac{1}{2} \times 6 \times 4 \times 4 = 0$$

$$R_B = \frac{48}{6} = 8\ kN$$

And $R_A = 12 - R_B = 12 - 8 = 4\ kN$



Here, $\frac{z}{x} = \frac{4}{6}$

$$z = \frac{4}{6}x$$

Now, $(S.F)_c = R_A - \frac{1}{2} \times x \times z$

Or $(S.F)_c = R_A - \frac{1}{2} \times x \times \frac{4}{6} \times x$

But $(S.F)_c = 0$

$$0 = 4 - \frac{1}{2} \times x^2 \times \frac{4}{6}$$

Or $x^2 = 12 = 3.4\ m$

Q.28 **A dry rock sample of diameter 50 mm and length 100 mm weighs 300 g. After saturating**

in brine solution of specific gravity 1.05, its weight increased to 330 g. The porosity of the rock sample in percentage is ________

Answer : 14.25 to 14.65.

Solution.

$$\text{Porosity} = \frac{Volume\ of\ void/liquid}{total\ volume\ of\ sample}$$

$$\text{Total volume of sample} = \pi\, r^2 l = 3.14 \times (25)^2 \times 100$$

$$= 196250\ mm^3$$

$$\text{Volume of void (liquid)} = \frac{mass\ of\ liquid}{density\ of\ liquid}$$

$$= \frac{mass\ of\ liquid}{specific\ gravity\ of\ liquid \times density\ of\ water}$$

$$= \frac{(330-300)\ g}{1.05 \times 1000\ kg/m^3}$$

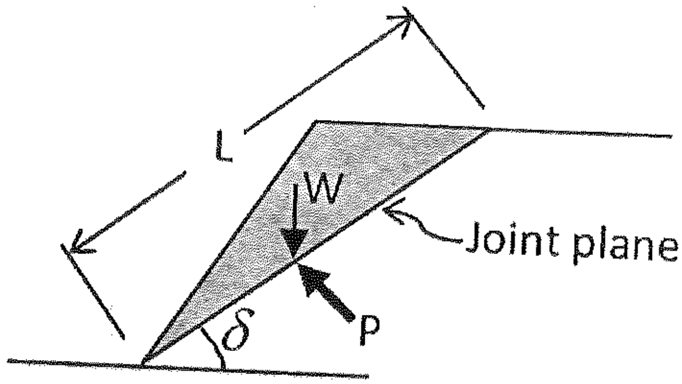
$$= \frac{30\ g \times m^3}{1.05 \times 1000\ kg}$$

Or $$= \frac{30 \times 1000^3}{1.05 \times 1000 \times 1000}\ mm^3$$

Hence, $$\text{Porosity} = \frac{30 \times 1000^3\ mm^3}{196250 \times 1.05 \times 1000 \times 1000\ mm^3} \times 100$$

$$= 14.55\ \%$$

Q.29 **A joint plane of length L and dip δ intersects the toe of a slope as shown in the figure. The weight of the shaded block is W. Uniform water pressure P acts normal to the joint plane. If the cohesion and angle of internal friction of the joint surface are c and ϕ respectively, then the expression for 'safety factor' of the shaded block is**



(A) $\frac{Lc+(W\sin\delta-LP)\tan\emptyset}{W\cos\delta}$

(B) $\frac{Lc+(W\cos\delta+LP)\tan\emptyset}{W\sin\delta}$

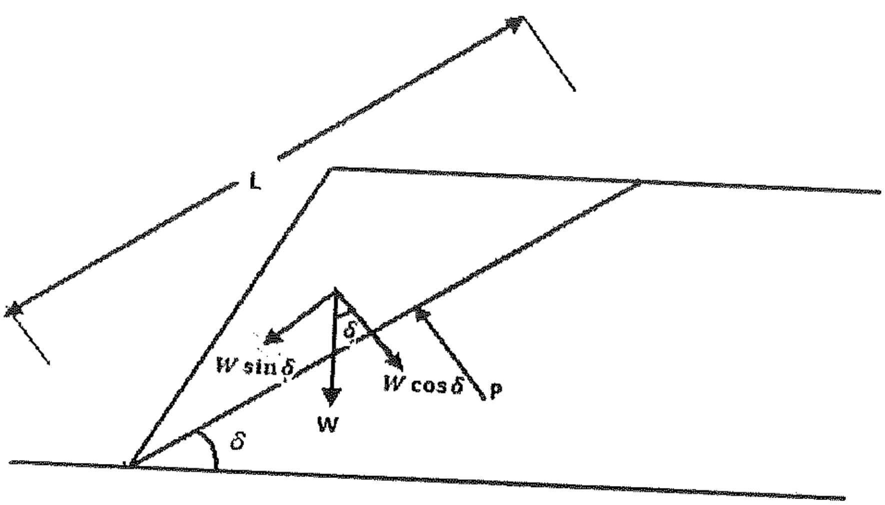
(C) $\frac{Lc+(W\cos\delta-LP)\tan\emptyset}{W\sin\delta}$

(D) $\frac{Lc+(W\sin\delta+LP)\tan\emptyset}{W\cos\delta}$

Answer (C)

solution .

Factor of safety = $\frac{Resisting\ force\ or\ shear\ force}{Driving\ force}$



From Mohr Coulomb failure criteria ,

Shear strength $= \tau = c + \sigma_n \tan\emptyset$

$[c - cohesion, \sigma_n - normal\ stress, \emptyset - Angle\ of\ internal\ \ friction]$

Therefore,

shear force $= \tau A = (c + \sigma_n \tan \emptyset).A = (c + N/A \tan \emptyset)A$

So, as per given condition :

Resisting force = shear force = $\left[c + \left(\frac{W \cos \delta}{L \times 1} - P\right) \tan \phi\right] L \times 1$

= Lc + (W.cosδ - PL) tanϕ

And driving force = w.sinδ

Hence, Factor of safety = $\frac{Lc + (W \cos \delta - LP) \tan \emptyset}{W \sin \delta}$

Q.30 **The lengths and standard errors of three sections AB, BC, and CD of a straight line AD are given below AB = 125.85 ± 0.021 m; BC= 205.72±0.029 m; CD = 246.21 ± 0.025 m. The standard error in total length AD in m is**

(A) ±0.0436 **(B) ±0.0350** **(C) ±0.0250** **(D) ±0.0019**

Answer (A)

Solution.

$e_{AB} = \pm 0.021$ m

$e_{BC} = \pm 0.029$ m

$e_{CD} = \pm 0.025$ m

The standard error in total length AD –

$$\pm e_s = \sqrt{(0.021)^2 + (0.029)^2 + (0.025)^2}$$

$$\pm e_s = 0.0436$$

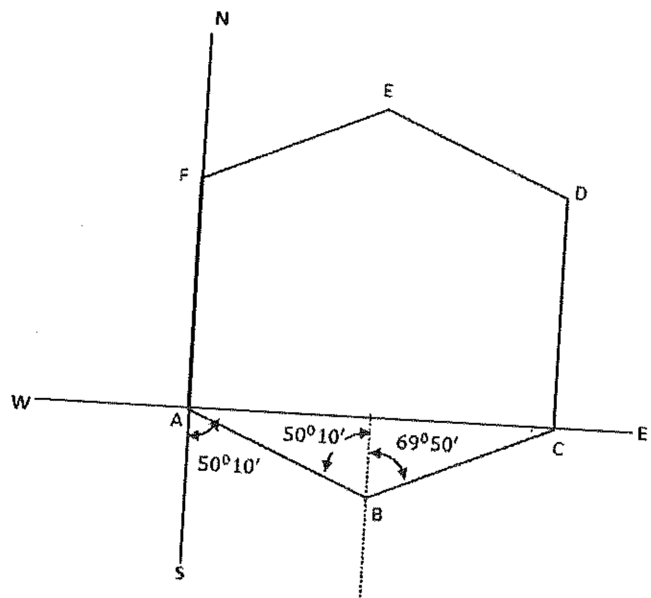
Or $e_s = \pm 0.0436$

Q.31 The bearing of side AB of a regular hexagon ABCDEF is $S50^\circ 10' E$. If the station C is easterly from the station B, the whole circle bearing of the side BC is

(A) $65^\circ 15' 25''$ (B) $69^\circ 50' 25''$ (C) $69^\circ 15' 25''$ (D) $69^\circ 50' 0''$

Answer (D)

Solution.



Hence, whole circle bearing of BC is $69^0 50' 0''$.

Note – In hexagonal ABCDEF

$$\angle A = \angle B = \angle C = \angle D = \angle E = \angle F = 120^0$$

Q.32 In a room-and-pillar stope, bench blasting is conducted using ANFO having density of 800 kg/m^3.The specific gravity of rock is 2.5, hole diameter is 100 mm and spacing to burden ratio is 1.3. The charge length of each blast hole is 80% of the hole length. For a desired powder factor of 0.48 kg/tonne, the spacing and burden of the blast pattern in m respectively are

(A) 2.0, 2.6 (B) 2.3, 1.8 (C) 5.2, 4.0 (D) 1.3,

Answer (B)

Solution.

Given that

ANFO density = 800 $\mathrm{kg} / \mathrm{m}^3$

Specific gravity of rock = 2.5

Hole dia. = 100 mm

$$\frac{Spacing\ (S)}{Burden\ (B)} = 1.3$$

Power factor = 0.48 kg/tone, and

Charge length of hole = 80% of hole length

Now,

$$\text{Powder factor} = \frac{Amount\ of\ explosive\ \ (Kg)}{Blasted\ material\ (tonne)}$$

$$0.48 = \frac{800 \times \pi \times (0.05)^2 \times charge\ length}{S \times B \times hole\ length \times 2.5}$$

$$0.48 = \frac{800 \times \pi \times (0.05)^2 \times charge\ length}{S \times B \times \frac{charge\ length}{0.8} \times 2.5}$$

$$0.48 = \frac{800 \times 3.14 \times (0.05)^2 \times 0.8}{1.3 \times B \times B \times 2.5}$$

$$B^2 = \frac{800 \times 3.14 \times (0.05)^2 \times 0.8}{0.48 \times 1.3 \times 2.5}$$

$\therefore$ B = 1.8m

and spacing = 1.3 x 1.6 = 2.3m

Q.33 Match the following for ore handling operations in an underground metal mine

Arrangement	Description
(P) Drawpoint	(I) arrangement that prevents oversized rock to pass
(Q) Ore pass	(II) a system of vertical or near vertical openings for transferring ore from a stope to a single delivery point
(R) Grizzly	(III) a place where ore can be loaded and removed
(S) Finger raise	(IV) a vertical or inclined opening used for transferring ore

(A) P-IV, Q-III, R-II, S-I

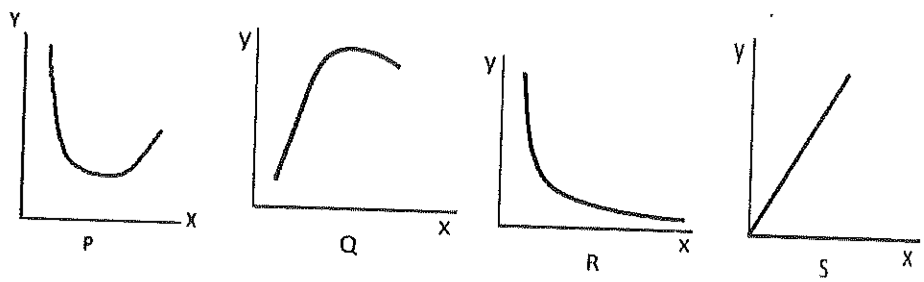
(B) P-III, Q-IV, R-I, S-II

(C) P-II, Q-IV, R-I, S-III

(D) P-III, Q-I, R-II, S-IV

Answer (B)

Q.34 The following characteristic curves (P, Q, R, S) pertain to rotary drilling in rock.



Title of the curve

I: Torque versus RPM

II: Rate of penetration versus uniaxial compressive strength of rock

III: Rate of penetration versus weight on bit

IV: Specific energy versus weight on bit

Match the curves with their titles

(A) P-III, Q-IV, R-II, S-I

(B) P-II, Q-IV, R-I, S-III

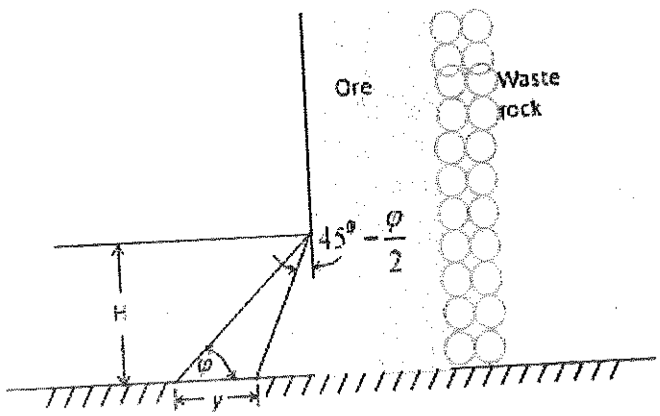
(C) P-IV, Q-III, R-II, S-I

(D) P-I, Q-III, R-II, S-IV

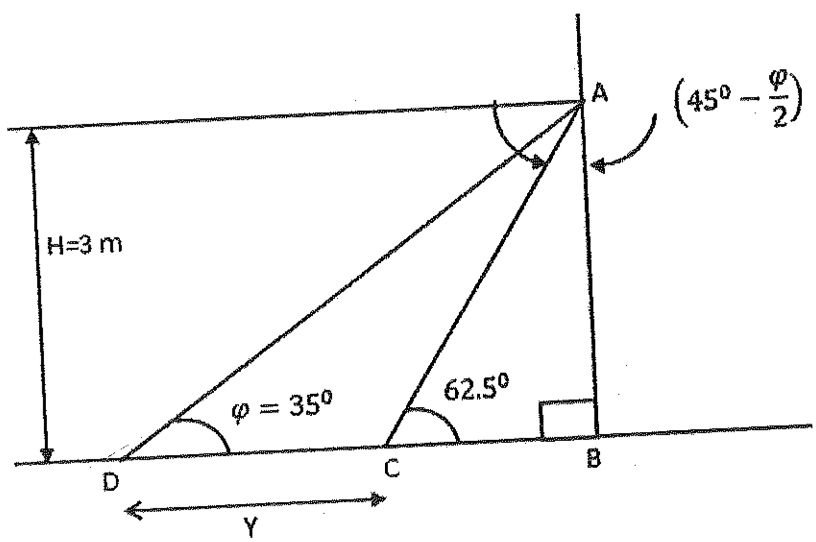
Answer (C)

Q.35 The height H of a drawpoint in a sublevel caving stope is 3.0 m. If the angle of repose (φ) of broken ore is 35^0, the digging depth y of the loader as shown in the figure in m is ______

Answer : 2.70 to 2.75.



Solution.



Given that, $\psi = 35^0$

Now,

$$\angle ACB = 90^0 - \left(45^o - \frac{\psi}{2}\right)$$

$$\angle ACB = 90^0 - \left(45^o - \frac{35^o}{2}\right)$$

$\angle ACB = 62.5^0$

In Δ ABC,

$$\tan 62.5^0 = \frac{3}{BC}$$

$\therefore \quad BC = 1.5617m$

Now, In Δ ABD

$$\tan 35^o = \frac{3}{Y+BC}$$

$$\therefore \quad Y + BC = \frac{3}{\tan 35^0}$$

$Y + 1.5617 = 4.28444$

$\therefore \quad Y = 2.72m$

Q.36 For an explosives company, the probability of producing a defective detonator is 0.02. The probability that a lot of 50 detonators produced by the company contains at most 2 defective detonators is __________.

Answer : 0.90 to 0.94.

Solution.

From Poisson distribution , Probability of observing x events in given interval is:-

$$P_x = \frac{e^{-m} m^x}{x!}, \qquad x = 0, 1, 2, 3, 4....$$

mean = m = np = 50 x 0.02 = 1

The probability that at most 2 defective detonators is

$$= \frac{e^{-m}.m^0}{0!} + \frac{m^1.e^{-m}}{1!} + \frac{m^2.e^{-m}}{2!}$$

$$= \frac{e^{-1}}{1} + \frac{1.e^{-1}}{1} + \frac{1^2.e^{-1}}{2!}$$

$$= \frac{1}{e} + \frac{1}{e} + \frac{1}{2e}$$

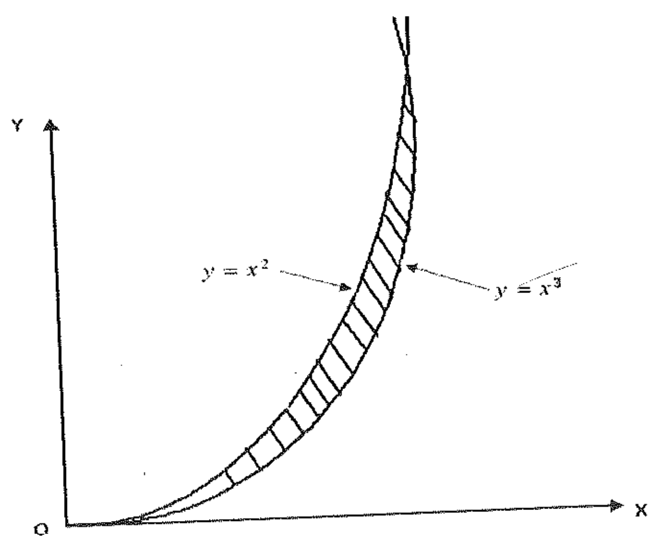
= 0.92

Q.37 The area enclosed by the curves $y = x^2$ and $y = x^3$ for $x \in [0, \infty]$ is

(A) 1/12 (B) 1/6 (C) 1/2 (D) 1

Answer (A)

Solution.



$y = x^2$ ....(i)

$y = x^3$ ....(ii)

from (i) & (ii)

$x^3 - x^2 = 0$

$x^2.(x-1) = 0$

Curves interest at (0, 0) and (1, 1) and we see that $x^3 < x^2$ in this interval.

Therefore, area is given by

$$\int_0^1 (x^2 - x^3)dx = \left(\frac{1}{3}x^3 - \frac{1}{4}x^4\right)_0^1$$

$$= \frac{1}{3} - \frac{1}{4} = \frac{1}{12}$$

Q.38 **The value of a, for which the function below is continuous at x = 1 is**

$$f(x) = \{2x + ax^2, \quad x \leq 1$$

$$\{4x + 3, \quad x > 1$$

(A) -5 **(B) 0** **(C) 5** **(D) 10**

Answer (C)

Solution.

Function is continuous at x = 1

if, $(2x + ax^2)_{at\ x=1} = (4x + 3)_{at\ x=1}$

$\therefore \quad 2 \times 1 + a \times 1^2 = 4 \times 1 + 3$

$$2 + a = 4 + 3$$

$\therefore \quad a = 7 - 2 = 5$

Q.39 The sum of the infinite series $a + ar + ar^2 + ar^3 + \ldots\ldots\ldots + ar^{n-1} + \ldots\ldots\ldots$ for $|r| < 1$ is

(A) a(1+r) (B) a(1−r) (C) a/(1+r) (D) a/(1−r)

Answer (D)

Solution.

The sum of n terms of a G.P is

$$S_n = \frac{a(1-r^n)}{1-r} \qquad \text{when } |r|<1$$

$$S_n = \frac{a}{1-r} - \frac{ar^n}{1-r}$$

Since, numerical value of r is less than 1. When value of n increases, then value of r^n decreases. When n tends to ∞, then value of r^n tends to 0. So sum of infinites series will be

$$S_\infty = \frac{a}{1-r}$$

Q.40 A centrifugal pump has a discharge rate of 2000 L of water per min against a total head of 200 m. If the pump efficiency is 75%, the input power to the pump in kW is

(A) 87.20 (B) 49.05 (C) 13.33 (D) 7.50

Answer (A)

Solution.

Given that,

Discharge rate = 2000 L/min

Total head = 200m

Pump efficiency = 75%

Input power to the pump = $\frac{P.Q \times 10^{-3}}{Pump\ efficiency}$ Kw

P = Pump pressure = ρgh = 1000 × 9.81 × 200 = 1962000 Pa

Q = 2000 l/min = $\frac{2000 \times 10^{-3}}{60}$ m^3/sec

Now,

Input power (P) to pump = $\frac{1962000 \times 2000 \times 10^{-3} \times 10^{-3}}{60 \times 0.75}$ Kw

$\therefore$ P = 87.2 KW

Q.41 **A dragline is required to remove 3,00,000 m^3 of rock per month on the bank volume basis.**

Consider the following data for the dragline operation.

Effective working hours per month = 450

Bucket fill factor = 0.8

Cycle time = 65 s

Swell factor of the rock = 1.25

The minimum bucket capacity of the dragline in m^3 is

(A) 7.70 **(B) 9.63** **(C) 12.04** **(D) 18.80**

Answer (D)

Solution.

Given that,

Dragline productivity per month = 3,00,000 m^3

effective working hour per month = 450 hour

Bucket fill factor = 0.8

Cycle time = 65 seconds

Swell factor of rock = 1.25

Here, dragline productivity per hour = $\frac{300000}{450}$

$= 666.67\, m^3$

Now, dragline productivity in m^3/h is given by

$$= \frac{3600 \times fill\ factor \times bucket\ capacity}{cycle\ time \times swell\ factor}$$

$$666.67 = \frac{3600 \times 0.8 \times bucket\ capacity}{65 \times 1.25}$$

$$\text{Bucket capacity} = \frac{666.67 \times 65 \times 1.25}{3600 \times 0.8}$$

$\therefore$ Bucket capacity = 18.80 m^3

Q.42 A direct rope haulage pulls 8 tubs loaded with coal through an incline of length 500 m having an inclination of 1 in 6. Consider the following additional data.

Capacity of tub = 1.0 tonne ,

Tare weight of tub = 500 kg

Hauling speed = 9 km per hour,

Coefficient of friction between wheel and rail = 1/60

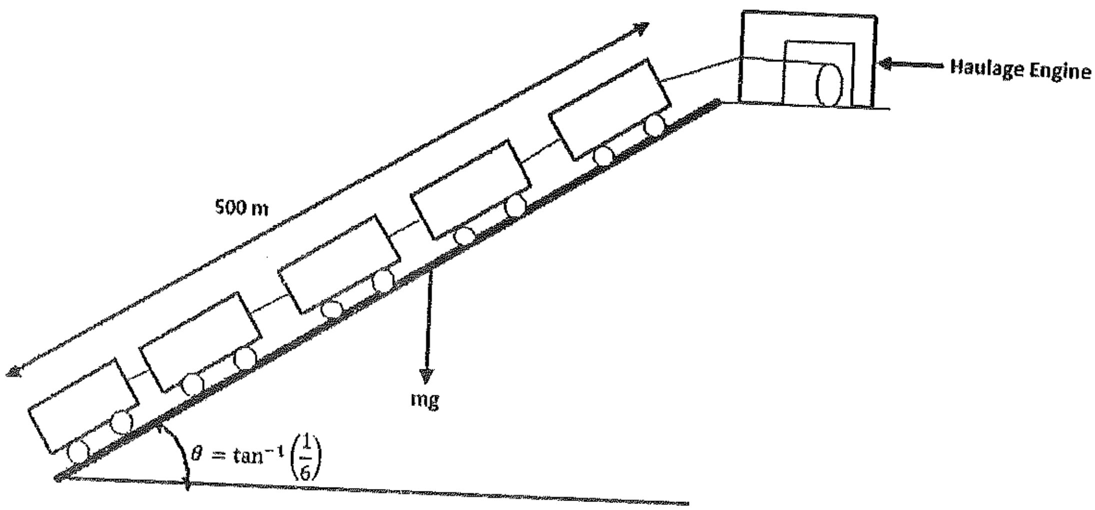
Coefficient of friction between rope and drum = 1/10 ,

Mass of rope per meter = 1.5 kg .

The minimum power required to haul the tubs in kW is

(A) 345.50 (B) 348.60 (C) 350.10 (D) 365.50

Solution.



Tractive force required = $G + g_1 + F + F_1$

G = Gravitational resistance of full tubes

G = (Weight of gross load per set) x g x inclination

$$G = 1500 \times 8 \times 9.81 \times \frac{1}{6} = 19620 \text{ N}$$

g_1 = Gradient Resistance of rope

g_1 = Rope length x rope mass per kg x g x inclination

$$g_1 = 500 \times 1.5 \times 9.81 \times \frac{1}{6} = 1226.25 \text{ N}$$

F = Friction of full tubs

F = (Weight of dross load per set) x g x coefficient of fiction between wheel and rail

$$F = 1500 \times 9.61 \times \frac{1}{6} = 1962\ N$$

F_1 = Friction of rope

F_1 = Rope length x rope mass per kg x g x coefficient of friction between rope and drum

$$F_1 = 500 \times 1.5 \times 9.81 \times \frac{1}{10} = 735.75N$$

Therefore,

Tractive force to overcome = 19620 + 1226.25 + 1962 + 735.75

= 23544 N

Hauling speed = 9 km/h = 2.5 m/sec

Minimum power (P) required to haul the tubs = Tractive force x speed

= 23544 x 2.5

= 58860 W

P = 58.86 KW

Q.43 **A coal mine receives two bids for purchase of a new dragline. The first bid quotes Rs. 150 crore as a price to be paid in full on delivery. The second bid quotes Rs. 180 crore as a price payable at the end of the third year after delivery. If the discount rate is 12%, the difference in NPV between the first and second bids in crore of rupees is ________**

Answer : 21.80 to 21.95.

Solution.

NPV is given by

$$NPV = \sum_{t=0}^{N} \frac{R_t}{(1+i)^t}$$

where, t – The time of cash flow

i – Discount rate

$R_t = the\ net\ cash\ flow$

For first bids –

t = 0, R = 150 Crore , i = 12% = 0.12

Here, $(NPV)_1 = \frac{150}{(1+0.12)^0} = 150$ Crore

for second bids

t = 3, R = 180 Crore, i = 12% = 0.12

Here, $(NPV)_2 = \frac{180}{(1+0.12)^0} = 128.12$ Crore

Therefore,

The difference in NPV between the first and second bids in Crore of rupees is =

150 – 128.2 = 21.8 Crores

Q.44 **Match the following in the context of underground mine environment**

Instrument	**Measuring parameter**
P. Haldane apparatus	**I. Humidity**
Q. Godbert-Greenwald apparatus	**II. Air velocity**
R. Hygrometer	**III. Mine air composition**
S. Anemometer	**IV. Ignition point temperature**

(A) P-II, Q-I, R-III, S-IV

(B) P-III, Q-IV, R-I, S-II

(C) P-IV, Q-II, R-III, S-I

(D) P-I, Q-III, R-IV, S-II

Answer (B)

Q.45 A mine airway having cross-section of 2.2 m × 2.2 m and length 500 m contains a bend. Given that the airway friction factor is 0.01 Ns^2m^{-4}, shock loss factor for the bend is 0.07 and density of air is 1.2 kg/m^3, the equivalent length of the airway in m is ________

Answer: 502 to 503.

Solution.

We have,

Pressure drop in mine airways:-

$$P = \frac{K\,S\,Q^2}{A^3} = \frac{K\,S\,V^2}{A}$$

When this equation is applied for bend then, becomes

$$\Delta P_s = \frac{K_b.S_b.V^2}{A}$$

where ΔP_S = Shock loss

K_b = Friction factor for bend

S_b = Surface area of bend

V = Velocity of air through bend

A = Cross seasonal area of bend

but $\Delta P_S = X.P_V$

$X \rightarrow$ Shock factor, P_V – Velocity pressure

Hence,

$$XP_V = \frac{K_b.S_b.V^2}{A}$$

$$X.\frac{1}{2}.\rho.v^2 = \frac{k_b S_b.V^2}{A}$$

Putting all values then,

$$0.07 \times \frac{1}{2} \times 1.2 = \frac{0.01 \times [2\,(2.2 + 2.2\,) \times l_b]}{2.2 \times 2.2} \quad [\ell_b - \text{length of bend}]$$

$$l_b = \frac{0.07 \times 1.2 \times 2.2 \times 2.2}{0.01 \times 8.8 \times 2}$$

ℓ_b = 2.31m

Equivalent length of airways = 500 + 2.31 = 502.31m

Q.46 **In order to estimate the NVP in a mine, measurements are made at the main fan as shown below.**

Fan speed (RPM)	**Fan drift pressure (Pa)**	**Fan quantity (m^3/s)**
800	**655**	**82.2**
950	**730**	**85.5**

The NVP in Pa is ____________

Answer : 258 to 263.

Solution.

We have

$$P = RQ^2$$

When, fan speed = 800 rpm, fan drift pressure = 655 Pa

Fan quantity = 82.2 m^3/S

$\therefore \quad N + 655 = R_1 \times (82.2)^2$

$$R_1 = \frac{N+655}{82.2^2} \quad \text{..............................(i)}$$

Where, N = N.V.P

when, Fan speed = 950 rpm, Fan drift pressure = 730 Pa

Fan quantity = 85.5 m^3/S

$N + 730 = R_2 \times (85.5)^2$

$$R_2 = \frac{N+730}{(85.5)^2} \quad \text{..................................(ii)}$$

In both, conditions resistance of mine will be same.

Hence, $R_1 = R_2$

$$\therefore \quad \frac{N+655}{82.2^2} = \frac{N+730}{85.5^2}$$

On solving , it is found that ,

N = 260.42 Pa.

Q.47 **The resistances of two splits A and B are 0.35 Ns^2m^{-8} and 0.05 Ns^2m^{-8} respectively. The combined resistance of the shafts and trunk airways is 0.4 Ns^2m^{-8}. A booster fan is planned to be installed in split A to increase the quantity flowing through it. Assuming that the surface fan continues to operate at a constant pressure of 1000 Pa, the critical pressure of the booster fan in Pa is ______**

Answer : 874 to 876.

Solution.

Given that,

Resistance of split A = $0.35\,Ns^2m^{-8}$

Resistance of split B = $0.05\,Ns^2m^{-8}$

Combined resistance of shaft and trunk airways = $0.4\,NS^2m^{-8}$

Surface fan pressure =1000 Pa

Now, critical pressure of booster fan is given by

$$P_c = \frac{P_f \times R_s}{R_t}$$

where P_f – surface fan pressure

R_s – Resistance of split in which booster fan installed

R_t – Resistance of shaft and trunk airways

$$\therefore \quad P_c = \frac{100 \times 0.35}{0.4}$$

$$\therefore \quad P_c = 875 \text{ Pa}$$

Q.48 **A pitot tube is inserted in a ventilation duct with the nose facing the air flow. A vertical U-tube manometer filled with alcohol (specific gravity 0.8) has been used for pressure measurements such that 10.2 mm is read as the total pressure and 8.8 mm as the static pressure. Given the density of air to be 1.2 kg/m³, the air velocity at the nose of the pilot tube in m/s is __________**

Answer : 4.20 to 4.35.

Solution.

We know,

$$P = \rho g h;$$

P – Pressure (Pa)

ρ – Density of fluid kg/m^3

g – Acceleration due to gravity (m/sec^2)

h – Head (height) (m)

Total pressure = $0.8 \times 1000 \times 9.81 \times \frac{10.2}{1000}$ Pa

= 80 Pa

Static pressure = $0.8 \times 1000 \times 9.81 \times \frac{8.8}{1000}$ Pa

= 69 Pa

$\because$ Total pressure = Static pressure + dynamic pressure

$\therefore$ Dynamic pressure = 80 – 69, Pa

= 11 Pa

Dynamic pressure $= \frac{1}{2}.\rho.v^2$

Where,

ρ – density of air

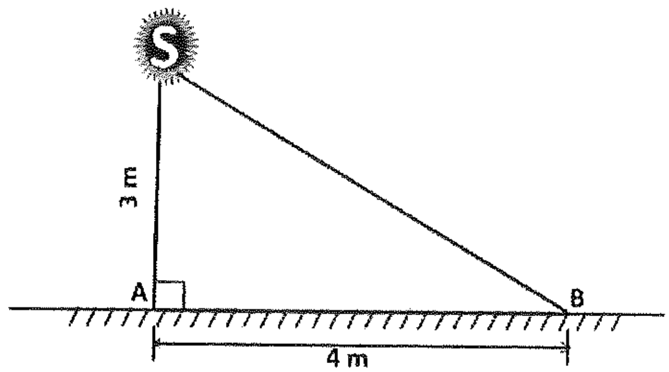
V – Velocity of air

$\therefore$ $11 = \frac{1}{2} \times 1.2 \times v^2$

Or $v = \sqrt{\frac{11 \times 2}{1.2}}$

v = 4.28 m/sec

Q.49

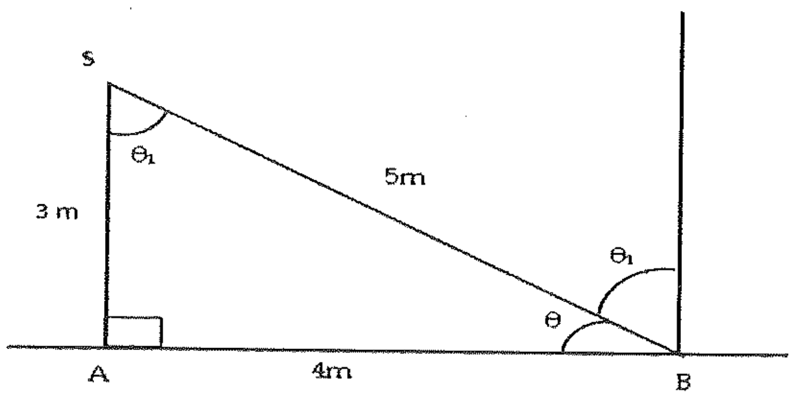


An illumination source S shown in the figure emits light equally in all directions. At a point A on the floor, the illuminance is 5.0 lux. The illuminance at point B on the floor in lux is ________

Answer : 1.0 to 1.2.

Solution.

Given figure



Now, $\theta = \tan^{-1}\frac{3}{4} = 36^0 52' 11.63''$

Hence $\theta_1 = 180^0 - \theta - 90^0 = 180^0 - 36^0\, 52'11.63'' - 90^0$

$= 53^0 7' 48.37''$

We have,

illumination of a surface (Meter candle) = (Candle of source) $\times \frac{\cos\theta_1}{(distance\ in\ m)^2}$

Here,

Candle of source = 5 x 3 x 3 = 45 lumens

$\cos\theta_1 = \cos 53^0 7' 48.37'' = 0.6$

Hence, illumination at point B will be-

$$= 45 \times \frac{0.6}{5^2}$$

= 1.08 lux.

Q. 50 **Two machines A and B while operating simultaneously produce a sound pressure level of 85 dBA at a point. When the machine A stops, the sound pressure level at that point reduces to 80 dBA. The sound pressure level at the same point due to machine A operating alone in dBA is**

(A) 70.0 **(B) 75.0** **(C) 80.0** **(D) 83.3**

Answer (D)

Solution.

We know,

$$L_{eq} = 10\log_{10}[\sum_{i=0}^{n} t_i \,.\, 10^{L_i/10}]$$

$$85 = 10\log_{10}\left[1\times10^{\frac{80}{10}} + 1\times10^{\frac{A}{10}}\right]$$

or $$8.5 = \log_{10}\left[10^8 + 10^{A/10}\right]$$

or $$10^{8.5} = 10^8 + 10^{A/10}$$

or $$10^{8.5} = 10^{8} + 10^{A/10}$$

$$\ln\left(10^{8.5} - 10^{8}\right) = \frac{A}{10}\ln(10)$$

A = 83.3 dB

Q. 51 **A waste water effluent has BOD_5 of 80 mg/L and the reaction rate constant is 0.16 per day. The ultimate BOD in mg/L is**

(A) 85 **(B) 100** **(C) 120** **(D) 145**

Answer (D)

Solution.

We have,

$$BOD_t = BOD_L \times \left(1 - e^{-Kt}\right)$$

where, BOD_t = BOD at any time t, mg/L = 80 mg/L

BOD_L = Ultimate BOD, mg/L

K = constant, reaction rate = 0.16

t = Time, day = 5 day

Hence, given BOD_t = 80 mg/L, k = 0.16, t = 5

$$80 = BOD_L \times \left(1 - e^{-5\times0.16}\right)$$

$$BOD_L = \frac{80}{1 - e^{-5\times0.16}} = 145 \text{ mg/L}$$

Q.52 **A series of tri-axial compression tests conducted on sandstone samples reveal the following relationship between major and minor principal stresses**

$\sigma_1 = 50 + 3\,\sigma_3$ [stresses are in MPa]

The cohesion in MPa and angle of internal friction in degrees of sandstone respectively are

(A) 14.43, 30.0 **(B) 14.43, 60.0** **(C) 0.21,73.9** **(D) 0.21, 16.1**

Answer (A)

Solution.

Given that

$$\sigma_1 = 50 + 3\,\sigma_3 \quad \text{.................................(i)}$$

and we have

$$\sigma_1 = C_o + \tan\beta\,.\,\sigma_3 \quad \text{.................................(ii)}$$

where, C_o – Uniaxial compressive strength

and angle β is constant

Comparing (i) and (ii), we found that,

C_o = 50 MPa and $\tan\beta = 3$

$$\tan\beta = \frac{1 + \sin\emptyset}{1 - \sin\emptyset}$$

where, ϕ – Angle of internal friction

$$\text{so,} \quad 3 = \frac{1+\sin\phi}{1-\sin\phi}$$

$\therefore$ $\phi = 30^0$,

and also we have,

$$\text{Uniaxial compressive strength } (\sigma_c) = \frac{2\,c\,\cos\emptyset}{1-\sin\emptyset}$$

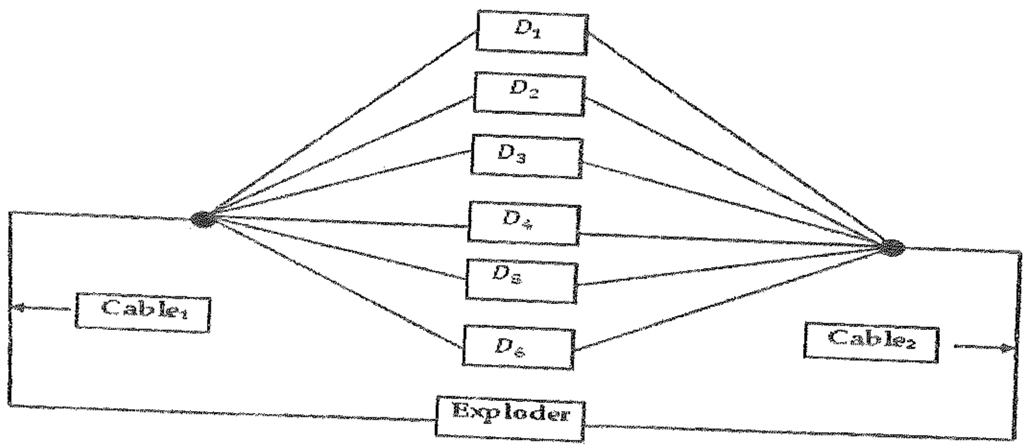
$$50 = \frac{2\,C\,\cos 30^0}{1-\sin 30^0}$$

$\therefore \quad C = 14.43 \text{ MPa}$

Q.53 **Six detonators each having resistance of 1.5 ohm are connected in parallel. A 15 V exploder is connected to the detonators by two single-core cables of resistance 3 ohm each. The current in the circuit in Ampere is __________**

Answer : 2.2 to 2.4.

Solution.



Given that, $D_1 = D_2 = D_3 = D_4 = D_5 = D_6 = 1.5 \text{ Ohm}$

and $Cable_1, = Cable_2, = 3$ Ohm

Now,

$$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \frac{1}{R_4} + \frac{1}{R_5} + \frac{1}{R_6}$$

$$\frac{1}{R}=\frac{1}{1.5}+\frac{1}{1.5}+\frac{1}{1.5}+\frac{1}{1.5}+\frac{1}{1.5}+\frac{1}{1.5}$$

$$R = \frac{1.5}{6}=0.25\,\text{Ohm}$$

Total resistance = 0.25 + 3 + 3 = 6.25 Ohm

Hence current in the circuit = $\frac{15}{6.25}$ = 2.4 Ampere

Q.54 **The failure and the repair rates of a shovel are 0.06 per hr and 0.04 per hr respectively. The availability of the shovel in percentage is ____________**

Answer : 40 to 40.

Solution.

$$\text{Availability} = \frac{Up\ time}{total\ time} = \frac{Up\ time}{Up\ time + down\ time}$$

$$= \frac{MTTF}{MTTF+MTTR}$$

Where,

MTTF → Mean time to failure

MTTR → Mean time to repair

Given that,

Failure rate = 0.06 hr^{-1}

Therefore , MTTF = $\frac{1}{0.06}=16.67\,\text{hr}$

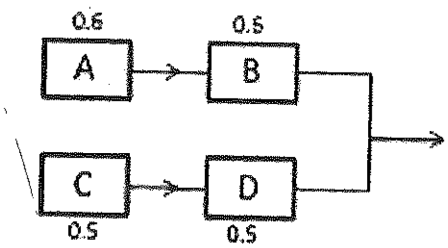
and Repair rate = 0.04 hr^{-1}

Therefore, MTTR = $\frac{1}{0.04}=25\,\text{hr}$

Hence shovel availability = $\dfrac{16.67}{16.67+25}=0.40$

= 40%

Q.55 **The individual reliability values of four sub-systems are given in the figure below. The reliability of the system is ________**



Answer : 0.50 to 0.55.

Solution.

Given,

Reliability values of sub systems A & B

$R_{AB} = 0.6 \times 0.6 = 0.36$

Reliability values of sub systems C & D

$R_{CD} = 0.5 \times 0.5 = 0.25$

Now, Reliability of the system is

$= 1-(1-0.36) \times (1-0.25)$

$= 0.52$

Q.56 **Choose the most appropriate word from the options given below to complete the following**

sentence.

A person suffering from Alzheimer's disease __________ short-term memory loss.

(A) experienced (B) has experienced

(C) is experiencing (D) experiences

Answer (D)

Q.57 Choose the most appropriate word from the options given below to complete the following sentence.

__________ is the key to their happiness; they are satisfied with what they have.

(A) Contentment (B) Ambition (C) Perseverance (D) Hunger

Answer (A)

Q.58 Which of the following options is the closest in meaning to the sentence below?

"As a woman, I have no country."

(A) Women have no country.

(B) Women are not citizens of any country.

(C) Women's solidarity knows no national boundaries.

(D) Women of all countries have equal legal rights.

Answer (C)

Q.59 In any given year, the probability of an earthquake greater than Magnitude 6 occurring in the Garhwal Himalayas is 0.04. The average time between successive occurrences of such earthquakes is ____ years.

Answer : 25 to 25.

Solution.

$P = 0.04 = \frac{4}{100}$

for 1 earthquake

$\frac{100}{4} p = 1$ earthquake

Reverse probability

= 25 years

Q.60 **The population of a new city is 5 million and is growing at 20% annually. How many years would it take to double at this growth rate?**

(A) 3-4 years **(B) 4-5 years** **(C) 5-6 years** **(D) 6-7 years**

Answer (A)

Solution.

We know, from compound interest

$$S = P(1 + i)^n$$

Where, S – value of n periods

P – Initial investments

i – Annual interest rate

n – Number of years

Here, we can apply such that,

S = 5 x 2 = 10, P = 5, i = 20% = 0.2

$$10 = 5(1 + 0.2)^n$$

$\frac{10}{5} = 1.2^n$

$2 = 1.2^n$

or $\log 2 = n \log 1.2$

$\therefore \quad n = \frac{\log 2}{\log 1.2} = 3 - 4$ years

Q.61 In a group of four children, Som is younger to Riaz. Shiv is elder to Ansu. Ansu is youngest in the group. Which of the following statements is/are required to find the eldest child in the group?

Statements

1. Shiv is younger to Riaz.
2. Shiv is elder to Som.

(A) Statement 1by itself determines the eldest child.

(B) Statement 2 by itself determines the eldest child.

(C) Statements 1 and 2 are both required to determine the eldest child.

(D) Statements 1 and 2 are not sufficient to determine the eldest child.

Answer (A)

Q.62 Moving into a world of big data will require us to change our thinking about the merits of exactitude. To apply the conventional mindset of measurement to the digital, connected world of the twenty-first century is to miss a crucial point. As mentioned earlier, the obsession with exactness is an artefact of the information-deprived analog era. When data was sparse, every data point was critical, and thus great care was taken to avoid letting any point bias the analysis.

From "BIG DATA" Viktor Mayer-Schonberger and Kenneth Cukier

The <u>main</u> point of the paragraph is:

(A) The twenty-first century is a digital world

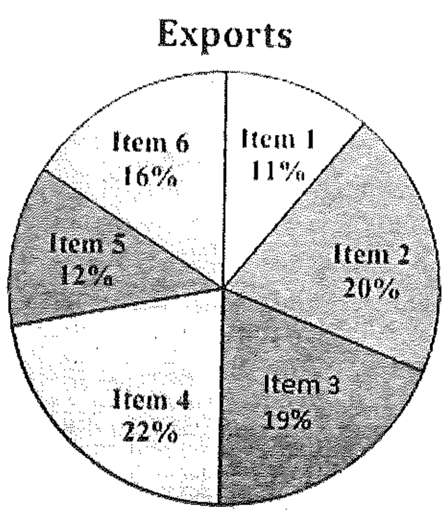
(B) Big data is obsessed with exactness

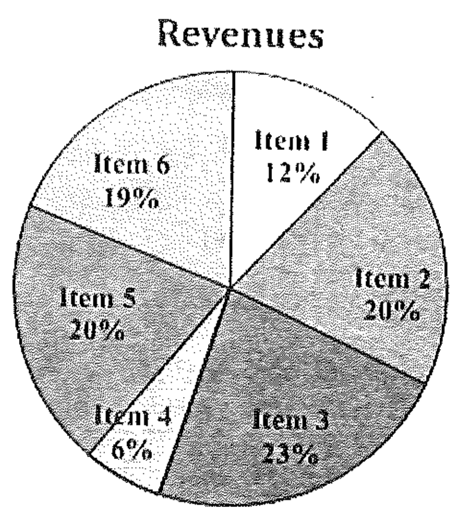
(C) Exactitude is not critical in dealing with big data

(D) Sparse data leads to a bias in the analysis

Answer (C)

Q.63 The total exports and revenues from the exports of a country are given in the two pie charts below. The pie chart for exports shows the quantity of each item as a percentage of the total quantity of exports. The pie chart for the revenues shows the percentage of the total revenue generated through export of each item. The total quantity of exports of all the items is 5 lakh tonnes and the total revenues are 250 crore rupees. What is the ratio of the revenue generated through export of Item 1 per kilogram to the revenue generated through export of Item 4 per kilogram?





(A) 1:2 (B) 2:1 (C) 1:4 (D) 4:1

Answer (D)

Solution.

For Item I,

Revenue = $\frac{12}{100} \times 250$ = 30 Crore rupees

Exports = $\frac{11}{100}$ x 5,00000 x 1000 = 55000000 Kg

Therefore,

Revenue generated through export of Item I, Per Kilograms = $\frac{30}{55000000}$

For Item IV,

Revenue = $\frac{6}{100}$ x 250 = 15 Crore rupees

Exports = $\frac{22}{100}$ x 5000000 x 1000 = 110000000 kg

Therefore,

Revenue generated through export of Item IV per kilogram = $\frac{15}{110000000}$

Hence,

Required ratio = $\dfrac{\frac{30}{55000000}}{\frac{15}{110000000}}$

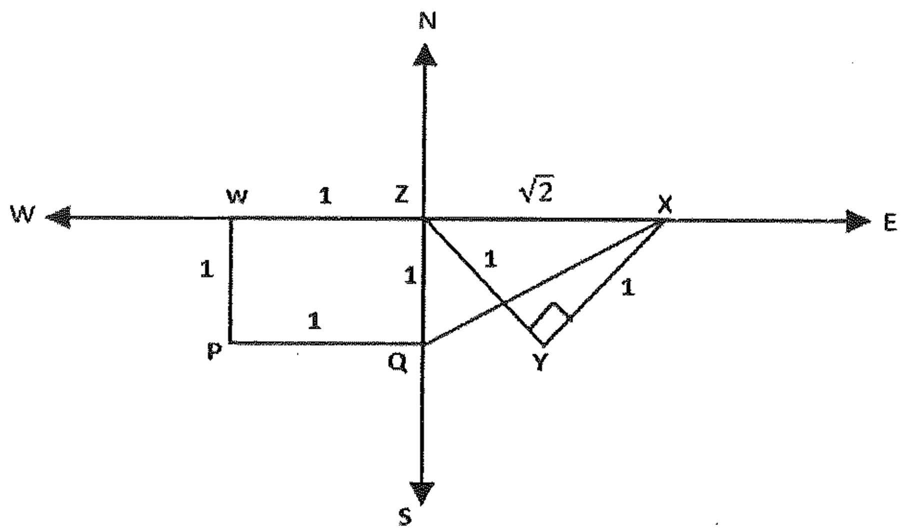
Required ratio = $\frac{4}{1}$ = 4:1

Q.64 **X is 1 km northeast of Y. Y is 1 km southeast of Z. W is 1 km west of Z. P is 1 km south of W. Q is 1 km east of P. What is the distance between X and Q in km?**

(A) 1 **(B) √2** **(C) √3** **(D) 2**

Answer (C)

Solution.



In ΔXYZ, [applying Pythagoras theorem]

We found that,

$$ZX = \sqrt{2}$$

Again in ΔXZQ [applying Pythagoras theorem]

$$XQ = \sqrt{(ZQ)^2 + (XZ)^2}$$

$$XQ = \sqrt{1^2 + \left(\sqrt{2}\right)^2}$$

$$XQ = \sqrt{3}\ \text{Km}$$

Q.65 10% of the population in a town is HIV^+. A new diagnostic kit for HIV detection is available; this kit correctly identifies HIV^+ individuals 95% of the time, and HIV^- individuals 89% of

the time. A particular patient is tested using this kit and is found to be positive. The probability that the individual is actually positive is _______

Answer : 0.48 to 0.49.

Solution.

Let total population = 100

$$\text{HIV + patient} = 100 \times \frac{10}{100} = 10$$

$$\text{HIV} - \text{patient} = 100 \times \frac{90}{100} = 90$$

For patient to be +ve = 10 x 0.95

and to be –ve = 10 x 0.5

For patient to be –ve = 90 x 0.89

and to be +ve = 90 x 0.11

Hence, probability that the individual is

$$\text{Actually positive} = \frac{10 \times 0.95}{10 \times 0.95 + 90 \times 0.11}$$

= 0.48